

MSE 360: Kinetic Processes in Materials

Homework #2

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. Answer true or false (and why):

- $Fe(\gamma) \rightarrow Fe(\alpha)$ is a homogeneous reaction.
- $\frac{dC_A}{dt} = -kC_A C_B$ is a first-order reaction rate law.
- The half-life of a first-order reaction is the time it takes for the reaction to go to half of its final value.

a) F ; α & γ are 2 different phases w/ distinct properties

b) F ; this is a 2nd order rxn that's 1st order wrt 2 reactants

c) F ; it is time to consume $\frac{1}{2}$ of reactant ($\frac{1}{2} C_0$)

4. In 2010, a stunning collection of Ice Age fossils, including several Woolly Mammoths, was found in an ancient high-altitude lake-bed near Snowmass, Colorado. Many of the fossils were "radiocarbon dead," meaning that they were so old nearly all traces of carbon-14 had decayed away. However, radiocarbon dating was successfully conducted on some of the "younger" mammoth fossils. If the half-life of ^{14}C is 5730 years:

- Calculate the reaction rate constant (yr^{-1}) for this radioactive decay.
- Calculate the age of the youngest mammoth fossils given that the ^{14}C concentration in these remains had decreased to 0.14 of the initial value at the time of sampling in 2010.
- Calculate the minimum age of the "radiocarbon dead" fossils given that ^{14}C concentration can be measured reliably down to 1% of the initial level.

$$a) \quad t_{1/2} = \frac{0.693}{k}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730 \text{ yr}} = \boxed{1.21 \times 10^{-4} \text{ yr}^{-1}}$$

$$b) \quad C_{14\text{C}} = C_{14\text{C}_0} \exp(-kt)$$

solve for t when $C_{14\text{C}} = 0.14 C_{14\text{C}_0}$

$$t = \frac{-\ln\left(\frac{C_{14\text{C}}}{C_{14\text{C}_0}}\right)}{k}$$

$$t = \frac{-\ln(0.14)}{1.21 \times 10^{-4} \text{ yr}^{-1}} = \boxed{16,000 \text{ years}}$$

c) solve for t when $C_{14\text{C}} = 0.01 C_{14\text{C}_0}$

$$t = \frac{-\ln(0.01)}{1.21 \times 10^{-4} \text{ yr}^{-1}} = \boxed{38,000 \text{ years}}$$